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Question 1:

A particle of mass m moves under the force F in $xy-plane$ such that at any time t its polar coordinates (r, θ) are given by (t, t^2). Find the transverse component of force.

Solution:

To find the transverse component of the force F acting on the particle moving in the $xy-plane$ we start with the given polar coordinates (r, θ)

$$r(t) = t$$

$$r(\theta) = t^2$$

The transverse component of the force is related to the angular component of the motion. First we need to express the velocity and acceleration in polar coordinates.

Velocity in Polar Coordinates

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The velocity $\vec{v}$ in polar coordinates (r, θ) is given by:

$$\vec{v} = \dot{r}\hat{e}_r + r\dot{\theta}\hat{e}_\theta$$

We need the time derivatives of r and θ

$$\dot{r} = \frac{d}{dt} r(t) = \frac{d}{dt} t = 1$$

$$\dot{\theta} = \frac{d}{dt} \theta(t) = \frac{d}{dt} t^2 = 2t$$

Thus, the velocity in polar coordinates is:

$$\vec{v} = 1.\hat{e}_r + t.2t\hat{e}_\theta$$

$$\vec{v} = \hat{e}_r + 2t^2\hat{e}_\theta$$

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$$\vec{v} = \hat{e}_r + 2t^2\hat{e}_\theta$$

Acceleration in Polar Coordinates

The acceleration $\vec{a}$ in polar coordinates is given by:

$$\vec{a} = (\ddot{r} - r\dot{\theta}^2)\hat{e}_r + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\hat{e}_\theta$$

$$\ddot{r} = \frac{d^2}{dt^2}r(t) = \frac{d^2}{dt^2}(t) = 0$$

$$\ddot{\theta} = \frac{d^2}{dt^2}\theta(t) = \frac{d^2}{dt^2}(t^2) = 2$$

Substituting these into the expressions for $\vec{a}$

$$\vec{a}_r = \ddot{r} - r\dot{\theta}^2 = 0 - t(2t)^2 = -4t^3$$

$$\vec{a}_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta} = 2t + 4t = 6t$$

So, the acceleration in polar coordinates is:

$$\vec{a} = -4t^3\hat{e}_r + 6t\hat{e}_\theta$$

Force in Polar Coordinates

Force in Polar Coordinates

The force $\vec{F}$ acting on the particle is related to the acceleration by Newton's second law

$$\vec{F} = m\vec{a}$$

Thus the components of the force are:

$$F_r = m(-4t^3)$$

$$F_\theta = m(6t)$$

Transverse Component of the Force

The transverse component of the force is the θ-component:

$$F_\theta = 6mt$$

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Transverse Component of the Force

The transverse component of the force is the θ-component:

$$F_\theta = 6mt$$

This is the force acting perpendicular to the radial direction in the particle's motion in the $xy-$ *plane*.

Question 2:

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A force of $(4\hat{\imath} + 5\hat{\jmath})\ N$ acts on a particle which displaces the particle with the velocity $(2\hat{\imath} + 7\hat{\jmath})$

Find the power applied on the particle.

Solution:

To find the power applied on the particle we use the formula:

$$\text{Power} = \text{Force} \cdot \text{Velocity} = \vec{F}.\vec{v}$$

$$P = \vec{F}.\vec{v}$$

First express the force and velocity vectors in terms of their components:

Given force $\vec{F} = (4\hat{\imath} + 5\hat{\jmath})N$

Given velocity $\vec{v} = 2\hat{\imath} + 7\hat{\jmath}$

Now let calculate the dot product of the force and velocity vectors:

$$\vec{F}.\vec{v} = (4\hat{\imath} + 5\hat{\jmath})(2\hat{\imath} + 7\hat{\jmath})$$

$$\vec{F}.\vec{v} = (2\hat{\imath}.4\hat{\imath} + 5\hat{\jmath}.7\hat{\jmath})$$

$$\vec{F}.\vec{v} = (2\hat{\imath}.4\hat{\imath} + 5\hat{\jmath}.7\hat{\jmath})$$

$$\therefore \hat{\imath}.\hat{\imath} = 1 = \hat{\jmath}.\hat{\jmath}$$

$$\vec{F}.\vec{v} = 8 + 35$$

$$\vec{F}.\vec{v} = 43$$

The dot product of the force and velocity vectors is 43 N m/s.

P = 43 N m/s

So the power applied on the particle is 43 Watts

Question 3:

Question 3:

Determine whether the force field $\overline{F} = (x^2y - z)\hat{\imath} + 2yz\hat{\jmath} + y^2\hat{k}$ is conservative or not?

Solution:

To determine whether the force field $\overline{F} = (x^2y - z)\hat{\imath} + 2yz\hat{\jmath} + y^2\hat{k}$ is conservative or not we can use a theorem from vector calculus: A vector field $\vec{F}$ is conservative if and only if its curl $\nabla \times \vec{F}$ is the zero vector

So let calculate the curl of $\vec{F}$

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2y - z & 2yz & y^2 \end{vmatrix}$$

Expanding this determinant:

$$\nabla \times \vec{F} = \left(\frac{\partial(y^2)}{\partial y} - \frac{\partial(2yz)}{\partial z}\right)\hat{\imath} - \left(\frac{\partial(y^2)}{\partial x} - \frac{\partial(x^2y - z)}{\partial z}\right)\hat{\jmath} + \left(\frac{\partial(2yz)}{\partial x} - \frac{\partial(x^2y - z)}{\partial y}\right)\hat{k}$$

$$\nabla \times \vec{F} = (0)\hat{\imath} + \hat{\jmath} + x^2\hat{k}$$

Since the curl of $\vec{F}$ is not the zero vector i.e. $\nabla \times \vec{F} \neq 0$ the force field $\vec{F}$ is not conservative.

Find the potential energy of a particle of mass $5.2\,kg$ which is at a height of $11m$ from the horizontal ground

Solution:

To find the potential energy of a particle at a height of 11 m above the ground we can use the formula for gravitational potential energy:

$$P.E = U = mgh$$

- U is the potential energy
- m is the mass of the particle $(5.2\,kg)$
- g is the acceleration due to gravity (approximately $9.8\,m/s^2$)
- h is the height above the reference point in this case the ground $(11m)$

Putting the values:

$$P.E = U = (5.2\,kg)(9.8\,m/s^2)(11m)$$

$$P.E = U = (5.2\,kg)(9.81\,m/s^2)(11m)$$

Putting the values:

$$P.E = U = (5.2\,kg)(9.8\,m/s^2)(11m)$$

$$P.E = U = (5.2\,kg)(9.81\,m/s^2)(11m)$$

$$P.E = U = 560.56\,J$$

So the potential energy of the particle at a height of 11 m from the ground is 560.56J

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